

```
$Assumptions = {E1 ∈ ℝ, E2 ∈ ℝ, E3 ∈ ℝ, E4 ∈ ℝ, p1 ∈ ℝ, p2 ∈ ℝ,  
[asunciones
```

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p3 ∈ ℝ, p4 ∈ ℝ, E1 > 0, E2 > 0, E3 > 0, E4 > 0, p1 > 0, p2 > 0, p3 > 0, p4 > 0};
```

```
Δ[vx_, vy_, vz_] :=
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```
Module[ {v =  $\sqrt{vx^2 + vy^2 + vz^2}$ ,  $\gamma = \frac{1}{\sqrt{1 - (vx^2 + vy^2 + vz^2)}}$ ,  $\gamma\text{bar} = \frac{1}{\sqrt{1 - (vx^2 + vy^2 + vz^2)}} - 1$ },  
[módulo
```

```
{ {γ, -γ vx, -γ vy, -γ vz},  
{ -γ vx,  $1 + \gamma\text{bar} \frac{vx^2}{v^2}$ ,  $\gamma\text{bar} \frac{vx vy}{v^2}$ ,  $\gamma\text{bar} \frac{vx vz}{v^2}$  },  
{ -γ vy,  $\gamma\text{bar} \frac{vx vy}{v^2}$ ,  $1 + \gamma\text{bar} \frac{vy^2}{v^2}$ ,  $\gamma\text{bar} \frac{vy vz}{v^2}$  },  
{ -γ vz,  $\gamma\text{bar} \frac{vx vz}{v^2}$ ,  $\gamma\text{bar} \frac{vy vz}{v^2}$ ,  $1 + \gamma\text{bar} \frac{vz^2}{v^2}$  } } ]
```

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In[47]:= η = { {1, 0, 0, 0}, {0, -1, 0, 0}, {0, 0, -1, 0}, {0, 0, 0, -1} };
```

```
P = {P1, P2, P3, P4};
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```
{P1, P2, P3, P4} =
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```
{ {E1, px1, py1, pz1},  
{E2, px2, py2, pz2},  
{E3, px3, py3, pz3},  
{E4, px4, py4, pz4} };
```

```
In[50]:= s1 = η . (P1 + P2) . (P1 + P2) // FullSimplify;  
[simplifica completa]
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s2 = η . (P3 + P4) . (P3 + P4) // FullSimplify;  
[simplifica completa]
```

```
t1 = η . (P3 - P1) . (P3 - P1) // FullSimplify;  
[simplifica completa]
```

```
t2 = η . (P4 - P2) . (P4 - P2) // FullSimplify;  
[simplifica completa]
```

```
u1 = η . (P3 - P2) . (P3 - P2) // FullSimplify;  
[simplifica completa]
```

```
u2 = η . (P4 - P1) . (P4 - P1) // FullSimplify;  
[simplifica completa]
```

```
$Assumptions = Join[$Assumptions, {α > 0, β > 0, mB ∈ ℝ}];  
[asunciones [junta [asunciones
```

```

In[64]:= {px1, px2, px3, px4} = {
    0,
    0,
    0,
    0};
{py1, py2, py3, py4} = {
    0,
    0,
    p3 Sin[α],
    -p4 Sin[β]};
{pz1, pz2, pz3, pz4} = {
    p1,
    -p2,
    -p3 Cos[α],
    p4 Cos[β]};
m1 = 0;
m2 = 0;
m3 = mB;
m4 = mB;

```

```

In[*]:= E1o = E1 /. Solve[P[[1]][[1]] ==  $\sqrt{m1^2 + P[[1]][[2]]^2 + P[[1]][[3]]^2 + P[[1]][[4]]^2}$ , E1][[1]][[1]] // Normal //
|resuelve |normal

FullSimplify;
|simplifica completamente

E2o = E2 /. Solve[P[[2]][[1]] ==  $\sqrt{m2^2 + P[[2]][[2]]^2 + P[[2]][[3]]^2 + P[[2]][[4]]^2}$ , E2][[1]][[1]] // Normal //
|resuelve |normal

FullSimplify;
|simplifica completamente

E3o = E3 /. Solve[P[[3]][[1]] ==  $\sqrt{m3^2 + P[[3]][[2]]^2 + P[[3]][[3]]^2 + P[[3]][[4]]^2}$ , E3][[1]][[1]] // Normal //
|resuelve |normal

FullSimplify;
|simplifica completamente

E4o = E4 /. Solve[P[[4]][[1]] ==  $\sqrt{m4^2 + P[[4]][[2]]^2 + P[[4]][[3]]^2 + P[[4]][[4]]^2}$ , E4][[1]][[1]] // Normal //
|resuelve |normal

FullSimplify;
|simplifica completamente

p1o = p1 /. Solve[P[[1]][[1]] ==  $\sqrt{m1^2 + P[[1]][[2]]^2 + P[[1]][[3]]^2 + P[[1]][[4]]^2}$ , p1][[1]][[1]] // Normal //
|resuelve |normal

FullSimplify;
|simplifica completamente

p2o = p2 /. Solve[P[[2]][[1]] ==  $\sqrt{m2^2 + P[[2]][[2]]^2 + P[[2]][[3]]^2 + P[[2]][[4]]^2}$ , p2][[1]][[1]] // Normal //
|resuelve |normal

FullSimplify;
|simplifica completamente

p3o = p3 /. Solve[P[[3]][[1]] ==  $\sqrt{m3^2 + P[[3]][[2]]^2 + P[[3]][[3]]^2 + P[[3]][[4]]^2}$ , p3][[1]][[1]] // Normal //
|resuelve |normal

FullSimplify;
|simplifica completamente

p4o = p4 /. Solve[P[[4]][[1]] ==  $\sqrt{m4^2 + P[[4]][[2]]^2 + P[[4]][[3]]^2 + P[[4]][[4]]^2}$ , p4][[1]][[1]] // Normal //
|resuelve |normal

FullSimplify;
|simplifica completamente

In[*]:= Solve[{p1^2 == p1o^2, p2^2 == p2o^2, p3^2 == p3o^2, p4^2 == p4o^2,
|resuelve
P[[1]][[1]] + P[[2]][[1]] == P[[3]][[1]] + P[[4]][[1]],
P[[1]][[2]] + P[[2]][[2]] == P[[3]][[2]] + P[[4]][[2]],
P[[1]][[3]] + P[[2]][[3]] == P[[3]][[3]] + P[[4]][[3]],
P[[1]][[4]] + P[[2]][[4]] == P[[3]][[4]] + P[[4]][[4]]
}, {E3, E4, p3, p4,  $\beta$ },  $\mathbb{R}$ ]

Out[*]=
$Aborted

```

(*Boost a frame donde positron está estático*)

$v_{CM} = v /. \text{Solve}[(\Delta[0, 0, v] \cdot P2 // \text{FullSimplify})[[4]] == 0, v][[1]][[1]]$
|resuelve |simplifica completamente

Out[*]=

$$-\frac{p2}{E2}$$